

CY 522: Computational Chemistry

Course Code	CY 522
Course Name	Computational Chemistry
L-T-P-C	3-0-0-3
Prerequisites	CY 521 Mathematics for Chemistry & CY 512 Advanced Quantum Chemistry
Intended for	UG/PG
Distribution	Core
Approval	9th Senate (May 2016)

Note: Course ran with CY 502; Course Code changed from CY 502 to CY 522 in 9th Senate

Course Contents

- **Introduction to Computational Chemistry**
Scope of computational chemistry, Numerical Methods (algorithms), Molecular Mechanics / Force Field Methods, molecular dynamics, Born-Oppenheimer approximation, potential energy surfaces, local and global minima, transition states. [16 Lectures]
- **Basics of Approximate methods**
Variational method, Hartree-Fock method. Molecular orbital theory, Slater determinants, anti-symmetry principle, restricted and unrestricted references, self-consistent-field (SCF) procedure. [6 Lectures]
- **Use of Approximate methods**
Basis sets, Hartree-Fock algorithm, Electronic spin degeneracy, evaluating the spin of Slater determinants, computational simplification using Group Theory, Semi-empirical methods, Geometry optimization. [6 Lectures]
- **Thermodynamic Properties**
Intrinsic reaction coordinate (IRC) analysis, Introduction to analytic gradient theory, Transition state theory, statistical mechanics, and thermodynamic properties. [8 Lectures]
- **Advanced Methods**
Electron correlation, Configuration interaction, Many-body perturbation theory. Comparing the performance of electronic structure theories. [6 Lectures]
Unit 6:
- **Hands on experience**
Hands on experience on the use of computational chemistry packages & understanding of experiments done in Laboratory courses. [0 Lecture]

Text Books

1. F. Jensen, **Introduction to Computational Chemistry**, 2nd Edition, John Wiley & Sons, 2007.
2. A. Szabo and N. S. Ostlund, **Modern Quantum Chemistry**, 2nd Edition, McGraw-Hill, 1989.

References

1. D. A. McQuarrie, **Quantum Chemistry**, Viva Books, 2011.
2. Ira N. Levine, **Quantum Chemistry**, Prentice-Hall, 1991.
3. F. A. Cotton, **Chemical Applications of Group Theory**, 3rd Edition, Wiley-Interscience, 1990.
4. C. J. Cramer, **Essentials of Computational Chemistry**, 2nd Edition, John Wiley & Sons, 2004.
5. A. R. Leach, **Molecular Modelling: Principle and Applications**, Prentice-Hall, 2001.
6. D. Young, **Computational Chemistry**, John Wiley & Sons, New York, 2001.
7. E. G. Lewar, **Computational Chemistry: Introduction to the Theory and Applications of Molecular and Quantum Mechanics**, Springer, 2003.